INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

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THE SITE

Location	CyrusOne PHX8, AZ -- station KPHX, America/Phoenix
Committed pair	OSM 1227682824 -> 977653858 CyrusOne PHX8 / CyrusOne PHX4
Facade gap	175.8 m between the two halls
Weather record	43,776 real hourly records from KPHX
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3200 C at 170 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-12-18 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 4 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. 9 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	4 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) 9 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

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hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	-	10.563	21.0	11.110	0.744
01	mechanical	yes	-	10.582	21.0	10.561	0.946
02	mechanical	yes	-	10.286	21.0	10.001	0.870
03	mechanical	no	refusal	10.038	21.0	9.440	0.803
04	mechanical	yes	switch budget	9.458	21.0	8.890	0.680
05	mechanical	yes	switch budget	8.898	21.0	7.780	0.804
06	mechanical	yes	switch budget	8.901	21.0	7.780	0.670
07	mechanical	no	refusal	10.037	21.0	8.330	0.976
08	mechanical	yes	-	12.232	21.0	11.110	1.178
09	mechanical	no	refusal	14.738	21.0	14.459	1.128
10	mechanical	yes	-	17.247	21.0	17.781	1.375
11	mechanical	yes	-	19.192	21.0	20.001	1.058

12	mechanical	no	dry-bulb	22.230	21.0	23.890	0.887
13	mechanical	no	dry-bulb	24.196	21.0	25.560	0.763
14	mechanical	no	dry-bulb	25.582	21.0	27.221	0.666
15	mechanical	no	dry-bulb	27.230	21.0	27.780	0.574
16	mechanical	no	dry-bulb	27.252	21.0	27.221	0.604
17	mechanical	no	dry-bulb	26.685	21.0	25.560	0.701
18	mechanical	no	dry-bulb	25.032	21.0	22.781	0.812
19	mechanical	no	refusal	22.640	21.0	19.066	1.020
20	FREE	yes	-	20.867	21.0	17.781	0.934
21	FREE	yes	-	19.197	21.0	17.221	0.898
22	FREE	yes	-	17.252	21.0	16.671	1.033
23	FREE	yes	-	16.417	21.0	16.111	0.881

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.563 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.582 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.286 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.458 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.898 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.901 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

08:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.232 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

10:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.247 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.192 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.230 C against a 21.0 C limit, so it fails by 1.230 C. It would take a limit 1.230 C higher, or a bound 1.230 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.196 C against a 21.0 C limit, so it fails by 3.196 C. It would take a limit 3.196 C higher, or a bound 3.196 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.582 C against a 21.0 C limit, so it fails by 4.582 C. It would take a limit 4.582 C higher, or a bound 4.582 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.230 C against a 21.0 C limit, so it fails by 6.230 C. It would take a limit 6.230 C higher, or a bound 6.230 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.252 C against a 21.0 C limit, so it fails by 6.252 C. It would take a limit 6.252 C higher, or a bound 6.252 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.685 C against a 21.0 C limit, so it fails by 5.685 C. It would take a limit 5.685 C higher, or a bound 5.685 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.032 C against a 21.0 C limit, so it fails by 4.032 C. It would take a limit 4.032 C higher, or a bound 4.032 C tighter, to change this hour.

19:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.867 C, which is 0.133 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.934 C, and the margin is measured, not chosen: 0.908 C of group-conditional forecast error for this hour of day, plus 0.0259 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.197 C, which is 1.803 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.898 C, and the margin is measured, not chosen: 0.873 C of group-conditional forecast error for this hour of day, plus 0.0259 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.252 C, which is 3.748 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.033 C, and the margin is measured, not chosen: 1.007 C of group-conditional forecast error for this hour of day, plus 0.0259 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.417 C, which is 4.583 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.881 C, and the margin is measured, not chosen: 0.855 C of group-conditional forecast error for this hour of day, plus 0.0259 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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